

Regulation of Gene Expression

Current 37-page faculty deck primary - official Week 4 objectives integrated - lac operon, multilevel eukaryotic control, epigenetics, imprinting, transcription factors, RNA interference, and X-inactivation

5 official objectives

Cis vs trans

Lac operon

Epigenetics

Imprinting

RNAi / XIST

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the source material for retrieval and application without replacing the deck.

Official Week 4 Objectives

1. Define gene expression; distinguish constitutive, facultative, regulated, temporal, and spatial expression; distinguish cis-acting regulatory elements from trans-acting factors and apply them to lac-operon regulation.
2. Identify the multiple levels at which eukaryotic gene expression can be regulated.
3. Describe the basic architecture of eukaryotic transcription factors, identify the four conserved DNA-binding motifs, and explain estrogen-receptor function.
4. Explain DNA methylation and histone modification in gene regulation and apply genomic imprinting to Prader-Willi, Angelman, and Beckwith-Wiedemann syndromes.
5. Describe regulatory roles of non-coding RNAs, including miRNA/siRNA RNA interference, and explain X-chromosome inactivation.

Constitutive

Housekeeping genes are expressed continuously because their products are needed broadly, such as core metabolic machinery.

Facultative / regulated

Expression is switched on or altered when a product is needed or in response to environmental, hormonal, or developmental signals.

Temporal

A gene is expressed at a particular time or developmental stage.

Spatial

A gene is expressed only in a particular tissue, cell type, or location.

Cis-acting elements are nucleotide sequences

They regulate nearby nucleic-acid sequences and do not need to encode a diffusible product. Examples in the deck include **promoters, enhancers, silencers/operators, and response elements**.

Trans-acting factors are diffusible products

Proteins or regulatory RNAs can move through the cell and interact with cis elements. Examples include transcription factors, activators, repressors, coactivators, corepressors, miRNAs, and lncRNAs.

Regulatory component	Core idea	High-yield relationship
Promoter	Defines where transcription begins	Sigma factor recognizes bacterial promoters; basal transcription factors assemble at eukaryotic core promoters
Enhancer	Binding site for activators	Raises transcription; can be far from the gene in eukaryotes
Silencer / operator	Binding site for repressors	Operator is the bacterial term emphasized in the deck
Response element	Sequence recognized by signal-responsive transcription factors	Examples: CRE, estrogen-response element, glucocorticoid-response element
Coactivator / corepressor	Adapter that integrates regulatory signals	Eukaryotic coactivators can bridge distant factors to basal transcription machinery

Fast discriminator: if the question points to a DNA/RNA sequence, think **cis**. If it points to a diffusible protein or regulatory RNA, think **trans**.

Deck pp. 8-10

Lac operon: lactose releases the brake; low glucose presses the accelerator LO 1

Bacterial transcription and translation are coupled, so regulation is primarily transcriptional. Functionally related genes can be clustered into an **operon** and transcribed as one **polycistronic mRNA**.

Lactose present → allolactose binds lac repressor → repressor leaves operator → transcription permitted

Glucose low → cAMP rises → cAMP-CRP/CAP binds activator site → RNA polymerase recruitment improves

Glucose	Lactose	Repressor	CRP/CAP	Lac transcription
High	Absent	Bound	Low activity	Off
High	Present	Released	Low activity	Low / basal
Low	Absent	Bound	Active	Off because the operator brake remains
Low	Present	Released	Active	Maximal

Do not collapse the two controls into one. Lactose controls the repressor; glucose availability is sensed through cAMP-CRP/CAP. Maximal expression needs lactose present *and* glucose low.

Deck pp. 11-12

Eukaryotes regulate expression at multiple checkpoints

LO 2

Genome / chromatin

Change whether regulatory DNA is physically accessible.

Transcription

Promoters, enhancers, silencers, response elements, and transcription factors determine RNA synthesis.

RNA processing / export

Splicing, transcript processing, stability, and nuclear export change the available mRNA pool.

Translation

Regulatory RNAs and translation-initiation pathways alter protein production from existing mRNA.

Post-translational

Modification, trafficking, activation, and degradation change protein abundance or activity.

Exam framing: eukaryotic regulation is not synonymous with transcriptional regulation. First identify *which level* is being altered.

Deck pp. 13-21

Epigenetics changes access to DNA without changing the DNA sequence

LO 4

Modification	Deck relationship	Expression effect
Histone acetylation by HAT	Neutralizes positive lysine charge and weakens histone-DNA interaction	Chromatin opens; expression generally rises
Histone deacetylation by HDAC	Restores tighter histone-DNA interaction	Chromatin compacts; expression generally falls
Histone methylation	Recruits regulatory proteins; effect depends on residue and methylation state	Can activate or repress - do not memorize it as always off
DNA CpG methylation	5-methylcytosine generated by DNMTs; promotes compact chromatin and reduced accessibility	Usually associated with reduced transcription

Trap: histone methylation is not the same rule as DNA methylation. The deck explicitly emphasizes that histone methylation can have different effects depending on the site.

DNA methylation is dynamic. Development includes waves of demethylation and remethylation, while imprinted loci preserve parent-of-origin information until germline reprogramming resets the imprint for the next generation.

Deck pp. 17-26

Genomic imprinting makes parental origin matter

LO 4

Imprinted allele = methylated / silenced allele in the deck's framework. If the only remaining copy of a gene is the normally imprinted copy, there may be no functional expression despite the gene still being physically present.

Condition	Missing active contribution	Classic mechanisms emphasized
Prader-Willi syndrome	Normally active paternal 15q11-q13 expression	Paternal deletion or maternal uniparental disomy
Angelman syndrome	Normally active maternal 15q11-q13 expression	Maternal deletion or paternal uniparental disomy
Beckwith-Wiedemann syndrome	Abnormal imprinting / parent-of-origin regulation at 11p15	Deck emphasizes paternal-only contribution / loss of normal maternal-region regulation as an overgrowth mechanism

PWS = paternal expression lost. AS = maternal expression lost. Do not answer imprinting questions by asking only which chromosome is deleted; ask which parent's normally active allele has been lost.

Deck pp. 27-28

Eukaryotic transcription factors are modular proteins

LO 3

Typical modules include a **DNA-binding domain**, a **transactivation / regulatory domain**, a nuclear localization signal, dimerization regions, and - for ligand-regulated nuclear receptors - a **ligand-binding domain**.

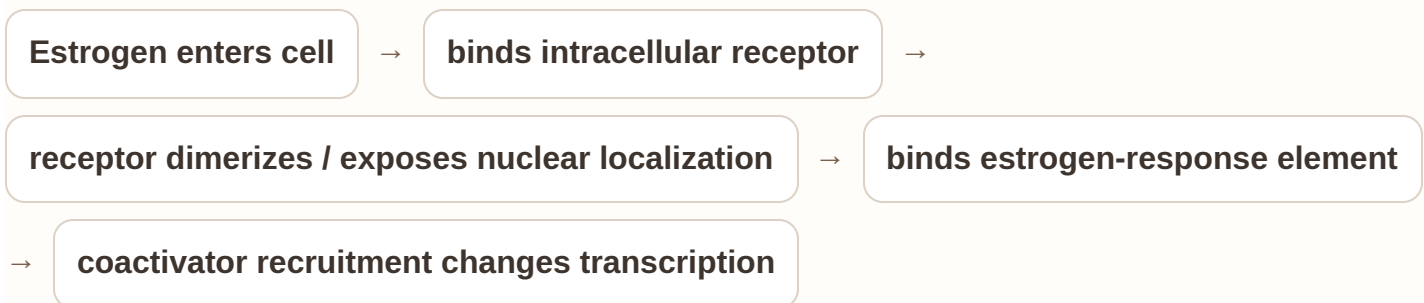
Conserved DNA-binding motif	Core exam association
Helix-turn-helix	Alpha-helical recognition element positions a helix in the major groove
Helix-loop-helix	Helical DNA binding plus dimerization
Leucine zipper	Leucines support dimerization; adjacent basic region can bind DNA
Zinc finger	Zinc-stabilized structure supports sequence-specific DNA binding; steroid receptors use zinc-finger DBDs

The deck's four motifs are explicitly: helix-turn-helix, helix-loop-helix, leucine zipper, and zinc finger.

Deck pp. 29-30

Estrogen receptor links a lipophilic signal directly to transcription

LO 3



The deck classifies estrogen receptor as a **Type I nuclear receptor**. In the absence of ligand, Type I receptors are cytoplasmic and associated with heat-shock proteins. Ligand binding changes receptor conformation, supports dimerization and nuclear entry, and allows response-element binding.

Tamoxifen: competes at the estrogen-receptor ligand-binding site but produces a receptor conformation that does not support the same coactivator interaction, thereby antagonizing estrogen-responsive transcription in the deck's model.

Deck pp. 33-34

Non-coding RNAs can silence genes after transcription

LO 5

Feature	miRNA	siRNA
Typical source	Encoded miRNA genes; Pol II transcripts	Long dsRNA, including exogenous / viral / synthetic sources
Processing	Drosha and Dicer pathways produce short guide RNAs	Dicer processes dsRNA into short guide RNAs
Target pairing	Often partial	Usually near-complete / complete
Target breadth	One miRNA may regulate many mRNAs	Generally more sequence-specific
Major outcome emphasized	Translational repression, enhanced degradation, or sequestration	Target mRNA degradation
Shared machinery	Both use RISC to direct silencing	

lncRNAs are >200 nucleotides and can regulate transcription, splicing, translation, chromatin, trafficking, and dosage compensation. **XIST** is the major lncRNA example in this lecture.

Deck pp. 35-36

X-inactivation is dosage compensation, not total silencing of every X-linked gene

LO 5

One X selected → XIST coats that chromosome →

DNA hypermethylation + histone deacetylation → stable heterochromatin / Barr body

- Selection is random between maternal and paternal X chromosomes in different embryonic cells.
- Descendants maintain the same inactive X, creating a **mosaic** pattern of X-linked gene expression.
- Some genes escape inactivation; pseudoautosomal-region genes are an important example in the deck.

Barr bodies = number of X chromosomes - 1 in the simple counting framework, but remember that genes escaping inactivation can still create dosage effects in sex-chromosome aneuploidy.

Exam traps

Protect the relationships most likely to be reversed

All LOs

Trap 1: cis = sequence; trans = diffusible factor.

Trap 2: maximal lac expression requires lactose present and glucose low.

Trap 3: HAT generally opens chromatin; HDAC closes it. Histone methylation is context-dependent.

Trap 4: PWS = loss of paternal expression; Angelman = loss of maternal expression.

Trap 5: steroid receptors are transcription factors with DNA-binding and ligand-binding domains; zinc fingers are the DBD motif emphasized.

Trap 6: miRNA and siRNA both use RISC, but target complementarity and dominant outcomes differ.

Trap 7: X-inactivation is incomplete; pseudoautosomal genes can escape.

30-SECOND FINISH

Cis is sequence; trans is diffusible. Lactose removes repression; low glucose activates CRP. Eukaryotes regulate chromatin, transcription, RNA, translation, and protein. HAT opens; DNA methylation usually silences. PWS loses paternal expression; AS loses maternal. TF motifs: HTH, HLH, leucine zipper, zinc finger. miRNA/siRNA use RISC. XIST creates the inactive X.